

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0125

Pittsfield Hydro 100 kV
May 16, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0125. The Interconnection Customer proposes a 505.6 MW solar photovoltaic facility interconnecting at the Pittsfield Hydro 100 kV substation in the Illinois area. The request was received on April 17, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 104 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$49,831,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Pittsfield Hydro-Santa Teresa 100.0 kV: 418.3% loading (130 MVA) following the outage of the Estill Springs-Ibm Bailey 500.0 kV; DFAX = 100.0%.

The Santa Teresa-Wommack 100.0 kV is overloaded to 280.3% of its 126 MVA emergency rating for the loss of the Santa Teresa-Pinhook 100.0 kV. The project contributes 68.5% DFAX (346.2 MW) to this overload.

Pinhook-Browns Ferry 100.0 kV: 271.6% loading (115 MVA) following the outage of the Santa Teresa-Wommack 100.0 kV; DFAX = 61.0%.

For the loss of Wommack-Santa Teresa 100.0 kV L, the Santa Teresa-Pinhook Substation 100.0 kV loads to 266.4% of its 126 MVA rating. Project DFAX: 61.0%.

The Crum Canyon-Browns Ferry 100.0 kV Line is overloaded to 202.3% of its 126 MVA emergency rating for the loss of the Santa Teresa-Wommack 100.0 kV. The project contributes 61.0% DFAX (308.6 MW) to this overload.

For the loss of Santa Teresa-Wommack 100.0 kV, the Army Depot-Zion Energy Center 100.0 kV loads to 176.5% of its 112 MVA rating. Project DFAX: 39.0%.

The Covanta Wallingford Energy-Zion Energy Ctr Substation 100.0 kV is overloaded to 134.2% of its 125 MVA emergency rating for the loss of the Santa Teresa-Wommack 100.0 kV. The project contributes 39.0% DFAX (197.0 MW) to this overload.

Guayama Pds-Davey Town 100.0 kV Circuit 2: 119.5% loading (145 MVA) following the outage of the Guayama Pds-Davey Town 100.0 kV Cir. 1; DFAX = 6.9%.

Plains Substation-Elk Grove 100.0 kV: 117.7% loading (191 MVA) following the outage of the EstillSprings 100/1kV XFMR; DFAX = 10.9%.

Davey Town-Guayama Pds 100.0 kV Line 1: 115.8% loading (149 MVA) following the outage of the Guayama Pds-Davey Town 100.0 kV Line 2; DFAX = 6.9%.

Medco-Covant Wallingford Energy 100.0 kV L: 105.7% loading (125 MVA) following the outage of the Santa Teresa-Wommack Substation 100.0 kV; DFAX = 33.1%.

The Dell-E 100.0 kV Line is overloaded to 104.5% of its 134 MVA emergency rating for the loss of the Potter County-Dell 100.0 kV. The project contributes 8.2% DFAX (41.3 MW) to this overload.

For the loss of Dell Substation-Guayama Pds 100.0 kV #2, the Dell-Guayama Pds 100.0 kV loads to 102.1% of its 141 MVA rating. Project DFAX: 6.9%.

For the loss of Dell-Guayama Pds 100.0 kV Ckt 1, the Dell-Guayama Pds 100.0 kV Cir. 2 loads to 102.1% of its 141 MVA rating. Project DFAX: 6.9%.

The Dell Substation-Potter County 100.0 kV is overloaded to 101.9% of its 128 MVA emergency rating for the loss of the E Substation-Dell 100.0 kV. The project contributes 7.7% DFAX (39.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pittsfield Hydro-Santa Teresa 100.0 kV	Rebuild line with higher-capacity conductor	\$3,526,000	100% (\$3,526,000)
Santa Teresa-Wommack 100.0 kV	Rebuild line with higher-capacity conductor	\$15,569,000	100% (\$15,569,000)
Browns Ferry-Pinhook 100.0 kV line	Rebuild line with higher-capacity conductor	\$1,758,000	100% (\$1,758,000)
Pinhook-Santa Teresa 100.0 kV	Rebuild line with higher-capacity conductor	\$6,412,000	100% (\$6,412,000)
Crum Canyon-Browns Ferry 100.0 kV	Rebuild line with higher-capacity conductor	\$5,556,000	100% (\$5,556,000)
Army Depot-Zion Energy Center 100.0 kV	Rebuild line with higher-capacity conductor	\$3,432,000	100% (\$3,432,000)
Covanta Wallingford Energy-Zion Energy Ctr 100.0 kV	Rebuild line with higher-capacity conductor	\$3,709,000	100% (\$3,709,000)
Guayama Pds-Davey Town 100.0 kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,779,000	100% (\$1,779,000)
Plains Substation-Elk Grove 100.0 kV	Rebuild line with higher-capacity conductor	\$2,890,000	100% (\$2,890,000)
Davey Town-Guayama Pds 100.0 kV Line 1	Rebuild line with higher-capacity conductor	\$1,821,000	100% (\$1,821,000)
Medco-Covant Wallingford Energy 100.0 kV L	Reconductor line	\$472,000	100% (\$472,000)
Dell-E 100.0 kV Line	Reconductor line	\$635,000	100% (\$635,000)
Dell-Guayama Pds 100.0 kV Line 1	Reconductor line	\$661,000	100% (\$661,000)
Dell-Guayama Pds 100.0 kV Cir. 2	Reconductor line	\$811,000	100% (\$811,000)
Dell Substation-Potter County 100.0 kV	Reconductor line	\$800,000	100% (\$800,000)
	Total allocated to project		\$49,831,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.